Fiscal Year:	FY 2005	**Task Last Updated:**	FY 10/26/2004
PI Name:	Takahashi, Fumiaki D.Eng.		
Project Title:	Flame Stability, Spread, and Suppression		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
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Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: Also affiliated with NASA Glenn Research Center		
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	Note: Original end date was 12/30/2007.		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Katta, Viswanath R. (Innovative Scientific Solutions, Inc.)		
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	This research relates to technology development efforts of both fire-extinguishing systems in human exploration of space and various combustion systems for terrestrial applications. The practical considerations are based on the fact that most uncontrolled fires and combustion systems are diffusion flame type combustion. Fire safety issues include flame stability, spread, and suppression phenomena aboard spacecraft in microgravity (ug) and on planetary surface bases in low gravity (the moon, 0.165g; Mars, 0.380g) as well as on the Earth (1g). As a result of flame-surface interactions, the diffusion flame often forms an edge, which may attach to burner walls, spread over condensed fuel surfaces, jump to another location through the fuel-air mixture formed, or extinguish by destabilization (blow-off). Furthermore, local extinction and re-ignition processes can occur in both fires and combustors as a result of interactions between the flame zone and vortices or fire-extinguishing agents. Flame spreading and secondary fires, where fires jump from one location to another, will lead an initially small ignition source to a		

Task Description:	dangerous established fire. In such circumstances, the dynamic behavior of edge diffusion flames may play a key role in a fate of fires. Because of various levels of buoyancy in low-gravity, fires in space behave differently from those on Earth. This research at NASA Glenn addresses edge diffusion flame stability, spread, and suppression, which is of essential importance in space-based fire safety and Earth-bound combustion systems. The structure of edge diffusion flames, stabilized on various geometries are studied experimentally and computationally. The experiments include both 1g and low-g tests in the 2.2 s Drop Tower at Glenn and the KC-135 Reduced-Gravity Aircraft. Diagnostics include video observations, the Mach-Zehnder Interferometry (MZI) for temperature measurements, and the particle image velocimetry (PIV) for velocity measurements. The fundamental aspect of the proposed program relates to recent advancement in numerical simulations. As a result of significant progresses in the development of detailed combustion reaction mechanisms and computer technologies over the last decade or two, it is now feasible to simulate various transient combustion phenomena in simple configurations (burner geometry, flow, and fuel) with confidence, leading to deeper understanding of physical and chemical unit processes taking place during the phenomena under investigation. Direct numerical simulations are performed by solving the time-dependent full Navier-Stokes equations along with energy- and species-conservation equations, including a detailed reaction mechanism for C1-C3 hydrocarbon combustion (33 species, 112 elementary steps) and radiative heat losses from CO2, H2O, CO, and CH4. The objectives of this research are: 1. To elucidate flame stability, spread, and suppression mechanisms of edge diffusion flames of gaseous, liquid, and solid fuels using various fire-extinguishing agents in 1g and low-g by revealing internal flame structure. The edge diffusion flame geometries to be used include axisymmetric fuel jets, candles, and cylindrical fuel surfaces in a low-speed coaxial oxidant stream in 1g and low-g with the oxygen concentration, oxidant velocity, pressure, fuel type, and agent type as independent variables. 2. To provide rigorous testing of the fidelity of the computational models, including detailed chemical reaction mechanisms and the radiative heat-transfer sub-models, in the simulations of the subject phenomena: unsteady flame stabilization, spread, and suppression, for various edge diffusion flame geometries, environments, fuels, and agents. This research is important in relation to fire safety in space exploration. The diffusion flame stability mechanism is more complicated and less understood than that of premixed flames because the reaction rates in diffusion flames are transport-limited and complex flame-structure dependent. A conventional view of diffusion flame stability in normal gravity, where a buoyancy-driven flow exists, or for forced-convection conditions assumes that there exists a small premixed flame at the flame base, which propagates against the mixture flow, thus preventing the flame from lifting. This research thus far has revealed that the speculation of premixed flame base formation is unrealistic, particularly in diffusion dominant micro gravity flames, and a newly conceived concept based on the reaction kernel replaces the conventional view. Direct numerical simulations with a detailed reaction mechanism and radiative heat losses provide precise flame structure information on radical species transport and reactions, which cannot be obtained by experiments or theories. Thus, this research contributes to expand our understanding of diffusion flame stabilization, particularly into a previously unstudied diffusion-dominant regime in microgravity. Furthermore, flame spread over condensed fuel surfaces and flame suppression have rarely been viewed from the point of stabilization and destabilization of edge diffusion flames. This research uniquely addresses to the edge diffusion flame phenomena in relation to the internal flame structure of the reaction kernel (peak reactivity spot) at the flame edge. The effects of various fire-extinguishing agents on the flame stability, spread, and suppression will be revealed on a step-by-step basis by systematic experiments and computations . Therefore, this research, which capitalizes the unique microgravity environment, benefits development efforts of fire-extinguishing systems to enable safe and productive human space explorations of the moon and Mars through fundamental understanding of the edge diffusion flame characteristics.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Underlying physics and chemistry of edge diffusion flames are relevant to the performance of Earth-bound combustion systems, including gas turbines, diesel engines, and industrial burners. Flame holding in combustors is necessary to achieve design performance and safe operation of the system. Therefore, this research is also beneficial for improvements and further developments of Earth-bound combustion technologies and certainly contributes to a progress in fundamental combustion science.
Task Progress:	The task progress has been published in two papers in the Proceedings of The Combustion Institute and a paper in the Proceedings of the Central States Sections/The Combustion Institute Meeting. The structure of axisymmetric laminar jet diffusion flames of ethane, ethylene, acetylene, and propane in quasi-quiescent air has been studied numerically in normal earth gravity (1g) and zero gravity (0g). The time-dependent full Navier-Stokes equations with buoyancy were solved using an implicit, third-order accurate numerical scheme, including a C3-chemistry model (32 species, 112 reactions) and an optically thin-media radiation model for heat losses. Observations of the flames were also made at the NASA Glenn 2.2-Second Drop Tower. For all cases of the fuels and gravity levels investigated, a peak reactivity spot, i.e., reaction kernel, was formed in the flame base, thereby holding a trailing diffusion flame. The location of the reaction kernel with respect to the burner rim depended inversely on the reaction-kernel reactivity or velocity. In the C2 and C3 hydrocarbon flames, the H2-O2 chain reactions were important at the reaction kernel, yet the CH3 + O --> CH2O + H reaction, a dominant contributor to the heat-release rate in methane flames studied previously, did not outweigh other exothermic reactions. Instead of the C1-route oxidation pathway in methane flames, the C2 and C3 hydrocarbon fuels dehydrogenated on the fuel side and acetylene was a major hydrocarbon fragment burning at the reaction kernel. The reaction-kernel correlations between the reactivity (the heat-release or oxygen-consumption rate) and the velocity, obtained previously for methane, were developed further for various fuels in more universal forms using variables related to local Damköhler numbers and Peclet numbers. Furthermore, direct numerical simulations with the C3-chemistry model have been performed to investigate the transient behavior and internal structure of flames propagating in an axisymmetric fuel jet of methane, ethane, ethylene, acetylene, or propane in 1g and 0g. The fuel issued from a 3-mm-i.d. tube into quasi-quiescent air for a fixed mixing time of 0.3 s before it was ignited along the centerline where the fuel-air mixture was at stoichiometry. The edge of the flame formed a vigorously burning peak reactivitiy spot, i.e., reaction kernel, and propagated through a flammable mixture layer, leaving behind a trailing diffusion flame. The reaction kernel broadened laterally across the flammable

	mixture layer and possessed characteristics of premixed flames in the direction of propagation and unique flame structure in the transverse direction. The reaction kernel grew wings on both fuel and air sides to form a triple-flame-like structure, particularly for ethylene and acetylene, whereas for alkanes, the fuel-rich wing tended to merge with the main diffusion flame zone, particularly methane. The topology of edge diffusion flames depend on the properties of fuels, particularly the rich flammability limit, and the mechanistic oxidation pathways. The transit velocity of edge diffusion flames, determined from a time series of calculated temperature field, equaled to the measured laminar flame speed of the stoichiometric fuel-air mixtures, available in the literature, independent of the gravity level.
Bibliography Type:	Description: (Last Updated: 06/13/2025)
Articles in Peer-reviewed Journals	Takahashi F, Katta VR. "Further studies of the reaction kernel structure and stabilization of jet diffusion flames." Proceedings of the Combustion Institute, 2005 Jan;30(1):383-90. http://dx.doi.org/10.1016/j.proci.2004.08.225 , Jan-2005
Articles in Peer-reviewed Journals	Takahashi F, Katta VR. "Structure of propagating edge diffusion flames in hydrocarbon fuel jets." Proceedings of the Combustion Institute, 2005 Jan;30(1):375-82. http://dx.doi.org/10.1016/j.proci.2004.08.227 , Jan-2005
Papers from Meeting Proceedings	Takahashi F, Katta VR. "Structure of propagating and attached hyrocarbon flames." Proceedings of the 2004 Technical Meeting of the Central States Section of the Combustion Institute, Austin, TX. March 2004. Proceedings of the 2004 Technical Meeting of the Central States Section of the Combustion Institute, March 2004. , Mar-2004